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In [10]: import pandas as pd
df=pd.read_csv('breast_cancer_clinical2.csv')
import scipy
from scipy.stats import shapiro
stat,p=shapiro(df['Age'])
print('Age的P值: ',p)
print('LN(-)和LN(+)两组的Age直方图:')
import matplotlib.pyplot as plt
plt.hist(df['Age'],bins=10)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
print('LN(-)和LN(+)两组的AgeQ-Q图: ')
import scipy.stats as stats
import matplotlib.pyplot as plt
stats.probplot(df['Age'],dist='norm',plot=plt)
plt.show()
print('age符合正态分布')
print('-----')
stat,p=shapiro(df['Tumor_Size'])
print('Tumor_Size的P值:',p)
print('LN(-)和LN(+)两组的Tumor_Size直方图:')
plt.hist(df['Tumor_Size'],bins=10)
plt.xlabel('Tumor_Size')
plt.ylabel('Frequency')
plt.show()
print('LN(-)和LN(+)两组的Tumor_Size Q-Q图: ')
stats.probplot(df['Tumor_Size'],dist='norm',plot=plt)
plt.show()
print('Tumor_Size符合正态分布')
print('-----')
stat,p=shapiro(df['Ki67'])
print('Ki67的P值:',p)
print('LN(-)和LN(+)两组的Ki67直方图:')
plt.hist(df['Ki67'],bins=10)
plt.xlabel('Ki67')
plt.ylabel('Frequency')
plt.show()
print('Ki67不符合正态分布')
print('-----')
print('LN(-)组: ')
group0=df[df['Lymph_Node']==0]
print(group0.describe())
print('LN(+)组: ')
group1=df[df['Lymph_Node']==1]
print(group1.describe())
print('Age和Tumor_Size的T_test:')
import scipy
from scipy.stats import ttest_ind
t,p=ttest_ind(group0['Age'],group1['Age'],equal_var=False)
print('Age的t值: ',t)
print('Age的p值: ',p)
t,p=ttest_ind(group0['Tumor_Size'],group1['Tumor_Size'],equal_var=False)
print('Tumor_Size的t值:',t)
print('Tumor_Size的p值: ',p)
table1=pd.DataFrame({
    'Variable':['Age','Tumor_Size'],
    'LN(-)':['43.75±4.86','2.2±0.43'],

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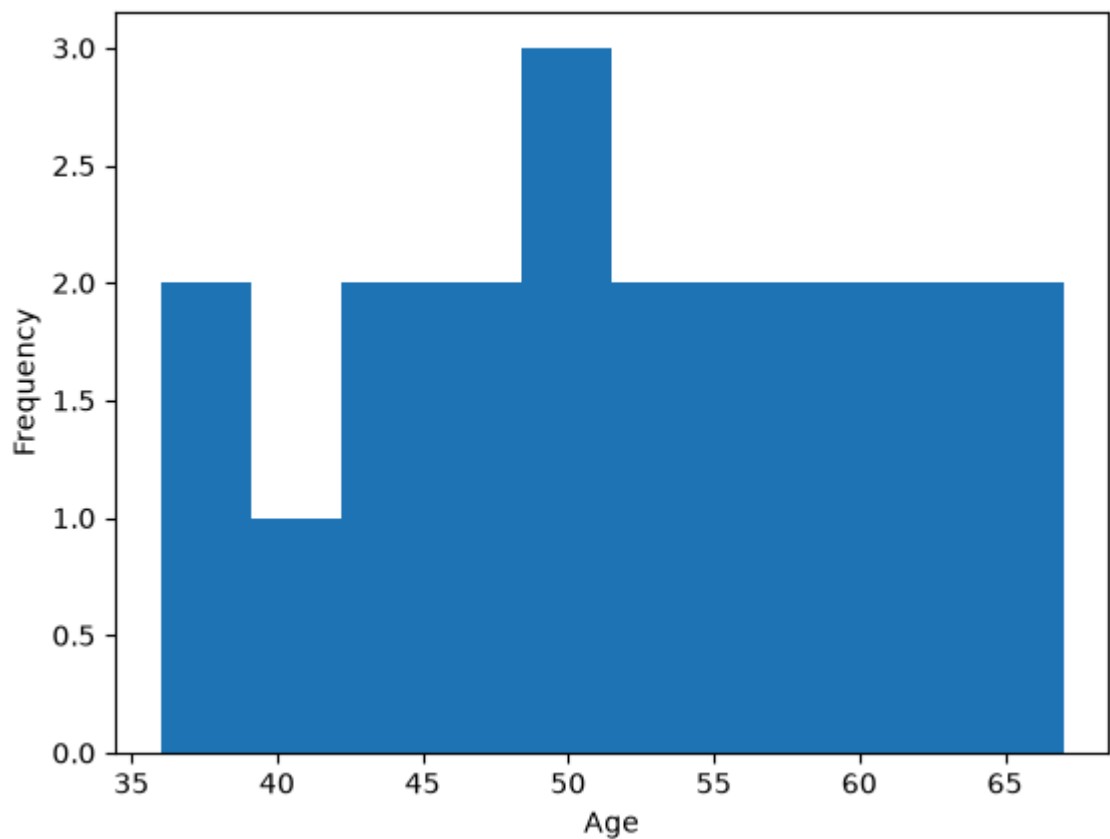
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    'LN(+)' : ['57.42±5.76', '3.86±0.69'],
    't value' : [-5.71, -6.57],
    'p value' : ['p<0.05', 'p<0.05']
})
print(table1)
table1.to_excel('T-test Results.xlsx')
print('-----')
print('Ki67的mann whitney U test:')
import scipy
from scipy.stats import mannwhitneyu
u,p=mannwhitneyu(group0['Ki67'],group1['Ki67'],alternative='two-sided')
print('Ki67的u值: ',u)
print('Ki67的p值: ',p)
table2=pd.DataFrame({
    'Variable':['Ki67'],
    'LN(-)' : ['20(13.75,52.5)'],
    'LN(+)' : ['35(18.75,70)'],
    'U' : [37],
    'P' : [0.41]
})
print(table2)
table2.to_excel('Mann-Whitney U Test Results.xlsx')

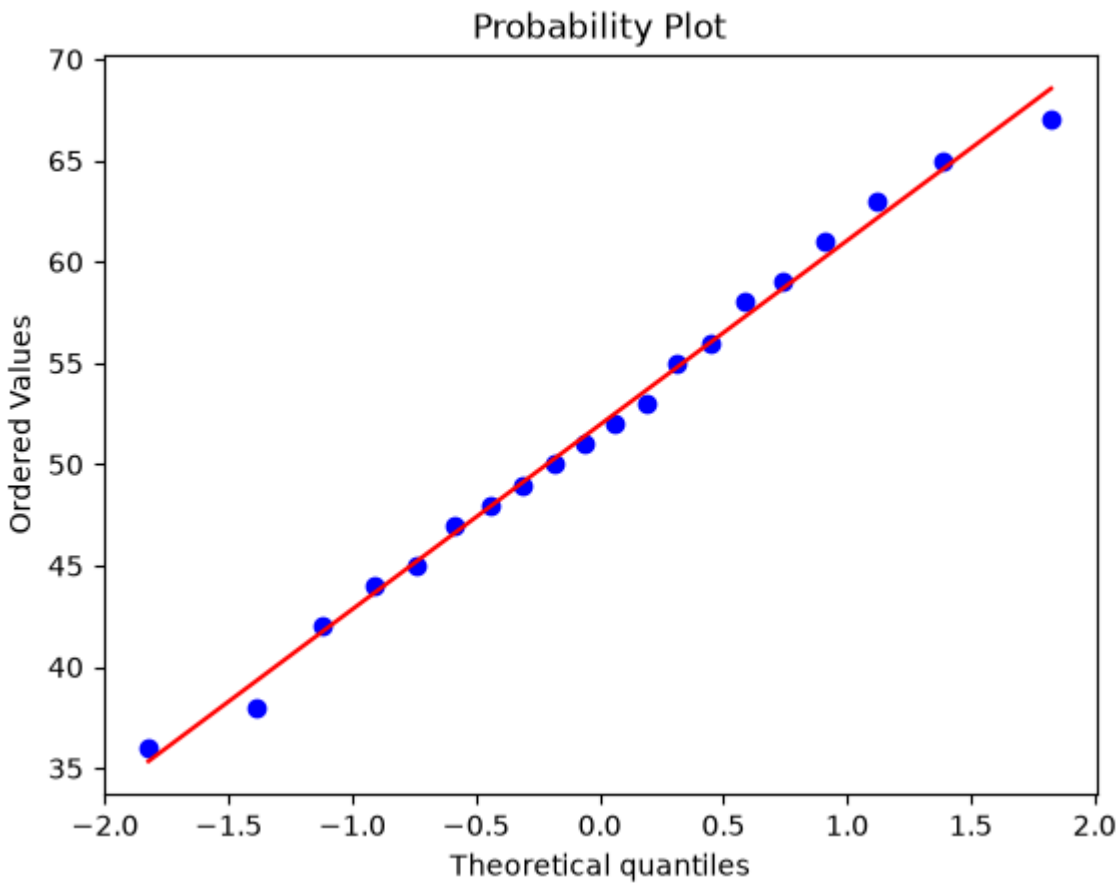
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Age的P值: 0.9788032525063932

LN(-)和LN(+)两组的Age直方图:



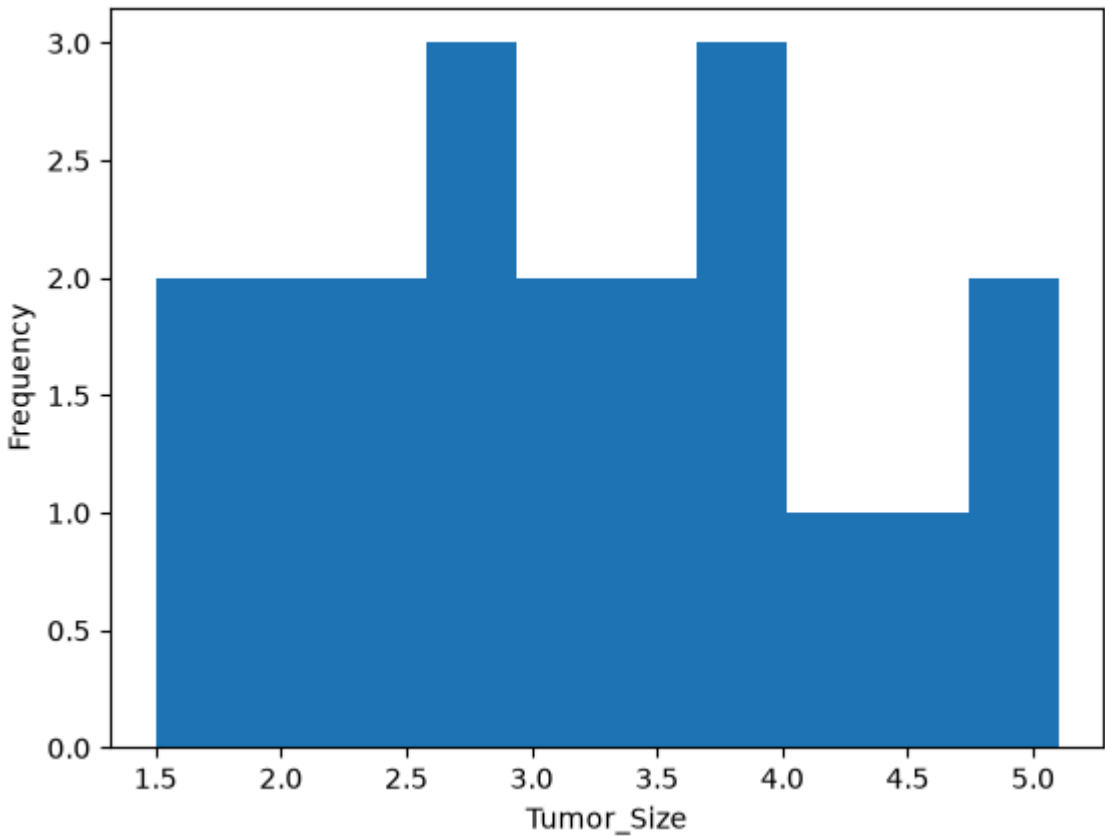
LN(-)和LN(+)两组的AgeQ-Q图:



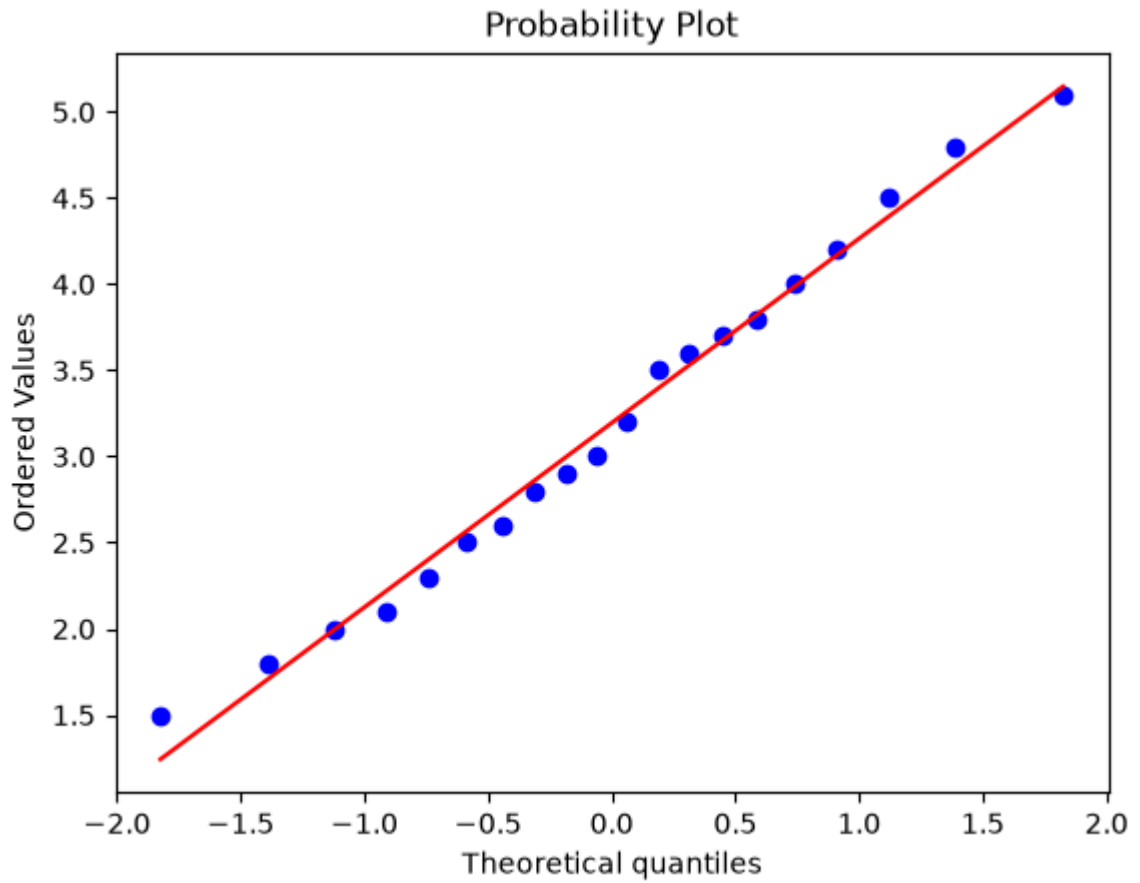
age符合正态分布

Tumor_Size的P值：0.9162922826518878

LN(-)和LN(+)两组的Tumor_Size直方图：



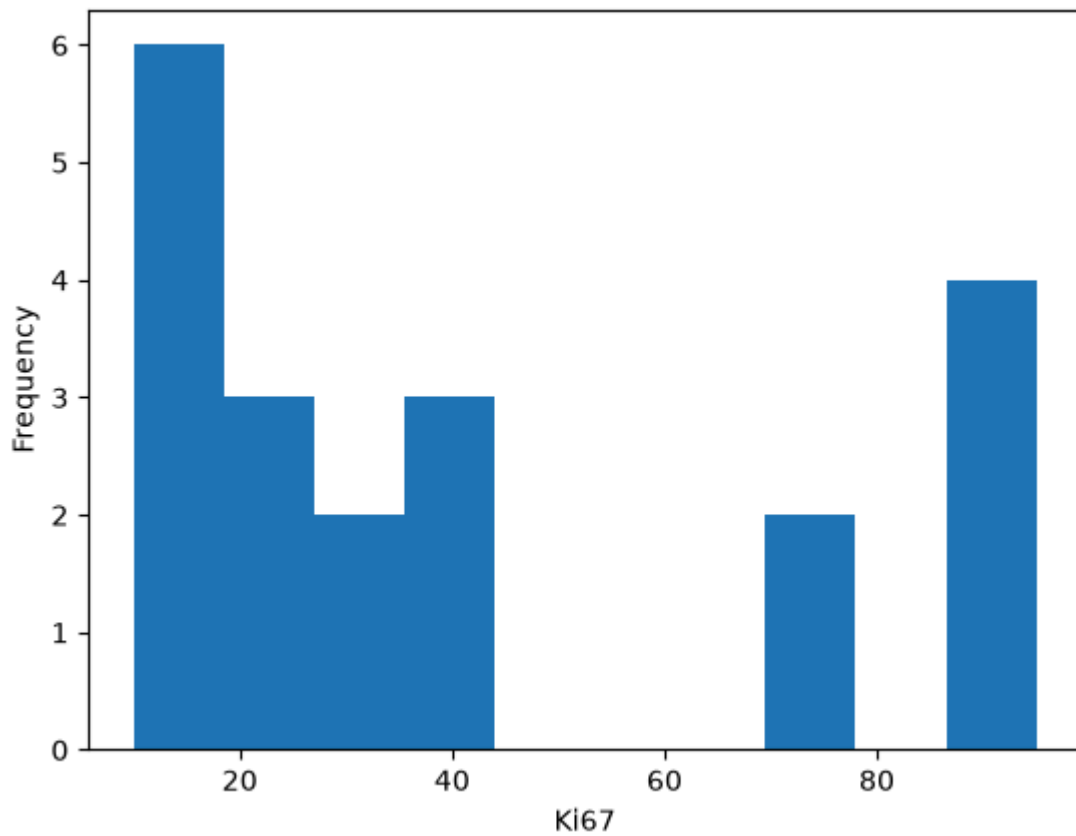
LN(-)和LN(+)两组的Tumor_Size Q-Q图：



Tumor_Size符合正态分布

Ki67的P值：0.002156689932776424

LN(-)和LN(+)两组的Ki67直方图：



Ki67不符合正态分布

LN(-)组:

	Age	Tumor_Size	Ki67	ER	PR	HER2	Lymph_Node
count	8.000000	8.000000	8.000000	8.0	8.000000	8.0	8.0
mean	43.750000	2.200000	36.875000	1.0	0.875000	0.0	0.0
std	4.862392	0.434248	34.115089	0.0	0.353553	0.0	0.0
min	36.000000	1.500000	10.000000	1.0	0.000000	0.0	0.0
25%	41.000000	1.950000	13.750000	1.0	1.000000	0.0	0.0
50%	44.500000	2.200000	20.000000	1.0	1.000000	0.0	0.0
75%	47.250000	2.525000	52.500000	1.0	1.000000	0.0	0.0
max	50.000000	2.800000	90.000000	1.0	1.000000	0.0	0.0

LN(+)组:

	Age	Tumor_Size	Ki67	ER	PR	HER2	\
count	12.000000	12.000000	12.000000	12.000000	12.000000	12.000000	
mean	57.416667	3.858333	44.166667	0.416667	0.166667	0.750000	
std	5.759709	0.693421	30.735923	0.514929	0.389249	0.452267	
min	49.000000	2.900000	10.000000	0.000000	0.000000	0.000000	
25%	52.750000	3.425000	18.750000	0.000000	0.000000	0.750000	
50%	57.000000	3.750000	35.000000	0.000000	0.000000	1.000000	
75%	61.500000	4.275000	70.000000	1.000000	0.000000	1.000000	
max	67.000000	5.100000	95.000000	1.000000	1.000000	1.000000	

	Lymph_Node
count	12.0
mean	1.0
std	0.0
min	1.0
25%	1.0
50%	1.0
75%	1.0
max	1.0

Age和Tumor_Size的T_test:

Age的t值: -5.7143816693673894

Age的p值: 2.6229721676823274e-05

Tumor_Size的t值: -6.573607448500683

Tumor_Size的p值: 3.5831577715728108e-06

	Variable	LN(-)	LN(+)	t value	p value
0	Age	43.75±4.86	57.42±5.76	-5.71	p<0.05
1	Tumor_Size	2.2±0.43	3.86±0.69	-6.57	p<0.05

Ki67的mann whitney U test:

Ki67的u值: 37.0

Ki67的p值: 0.4143554061444077

	Variable	LN(-)	LN(+)	U	P
0	Ki67	20(13.75,52.5)	35(18.75,70)	37	0.41

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